

Module 4A – Gravity, the Solar System, and Kepler’s Laws



In Module 2A, we learned about **Galileo’s Law of Falling Bodies**, which states that

Excluding aerodynamic factors, objects of dissimilar mass, released from the same height off the ground, will hit the ground in the same amount of time (location dependent).

The reason that this law is true is that **freely falling objects**,

objects falling under the influence of the force of gravity only,

fall with a constant (or uniform) acceleration.

The term **acceleration** means

the time rate of change of an object’s velocity,

and if an acceleration is going to be constant, its

direction must not change, and its speed must change linearly.

For an object freely falling (on Earth), its direction is certainly not changing, and its speed is increasing approximately 9.8 meters per second every second.

If an object is let go from a relatively large height (so it will fall for a while), absent air friction, its speed at

time t equal to 0 seconds is 0 meters per second,

at time t equal to 1 second, the object’s downward speed is 9.8 meters per second,

at time t equal to 2 seconds, the object’s downward speed is 19.6 meters per second,

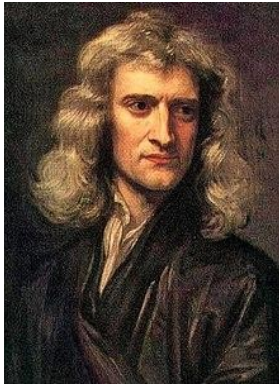
and so on.

But if freely falling objects uniformly accelerate, why is that?

In order to explain this, we must first explore **Newton's Second Law of Motion**, which states that

when a net external force $\vec{F}_{net,ext}$ acts on an object, the object accelerates ($\vec{a}$), its velocity changes.

By Newton, we mean the inimitable Sir Isaac Newton, one of the top two physical scientists of all time. His biography can be found [here](#).



Mathematically, we write **Newton's Second Law of Motion** as

$$\vec{F}_{net,ext} = m \vec{a}, \quad (3.1)$$

where m is the mass of the object accelerating.

The net external force is determined by adding up all the external forces acting on the body:

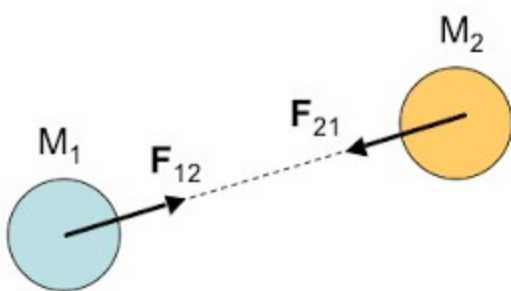
$$\sum_{i=1}^n \vec{F}_{ext,i} = \vec{F}_{net,ext} \quad (3.2)$$

For a freely falling object, the only external force acting on the object is the force of gravity, its **weight**, W . We write (from the Second Law, and without vector notation):

$$W = mg \quad (3.3)$$

Where g ("little g ") is the acceleration due to gravity.

The force of gravity, W , that produces that acceleration g comes from the Earth itself, and is described by **Newton's Law of Gravity**, which states that



There exists a universally attractive force between any two masses called gravity. The magnitude of this force is proportional to the product of the two masses, and inversely proportional to the square of the distance between their centers.

Mathematically, we write:

$$\vec{F}_{12} = (G M_1 M_2 / r^2) \hat{r}_{12} \quad (3.4)$$

Where $\vec{F}_{12}$ is the gravitational force on mass M_1 due to mass M_2 , r is the distance between the centers of M_1 and M_2 , G is the universal gravity constant equal to $6.67 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$, and $\hat{r}_{12}$ is the unit vector that points from M_1 to M_2 .

From formula (3.4), we see that your personal weight is

$$W = (G M_{Earth} m) / r^2 \quad (3.5)$$

where m is your personal mass, M_{Earth} is equal to the mass of the Earth, G is the universal gravity constant, and r is the distance between the center of the Earth and your center (your belly button).

But your personal weight (from the Second Law) is also

$$W = mg \quad (3.3)$$

We conclude that the acceleration due to gravity (on you or any object) is

$$g = G M_{Earth} / r^2 \quad (3.6)$$

Since the average radius of “spherical” Earth is 6,370 km or 3,959 mi, the few feet up from Earth’s surface to your belly button is extremely small compared to Earth’s radius, leading us to conclude that

$$r \cong R_{Earth}$$

and so, near Earth’s surface,

$$g = G M_{Earth} / R_{Earth}^2$$

which is a constant.

This is why

freely falling bodies near the surface of the Earth fall with the same acceleration, g , a constant.

Clearly, the analysis above shows that the constancy of the acceleration due to gravity (g) is only an approximation. For significant heights off the surface of the Earth, g declines significantly. Using formula (3.6), we see that for the International Space Station, which is 250 miles off the surface of the Earth,

$$\begin{aligned} g_{\text{Space Station}} &= G M_{Earth} / r^2 \\ &= 6.67 (5.98) \times 10^{-11+24} / 6.77^2 \times 10^{6+6} \\ &= 8.697 \text{ m/sec}^2 \end{aligned}$$

a reduction in value of about 11%, compared to the g at the surface of the Earth.

Little g (for Earth) is not even zero for a distance equal to how far the Moon is away. Using formulas (3.6) and (3.5) respectively, and knowing that

the mass of the Moon (M_{Moon}) is 7.35×10^{22} kg,

we see that

$$g_{\text{Earth at Moon's Distance}} = 2.71 \times 10^{-3} \text{ m/sec}^2$$

$$W_{\text{Moon, due to Earth's Gravity}} = 1.988 \times 10^{20} \text{ Newtons,}$$

where a Newton of force is about a quarter of a pound (lb) of weight.

This weight (the force on the Moon due to the Earth) is a very large force, and keeps the Moon in orbit around the Earth.

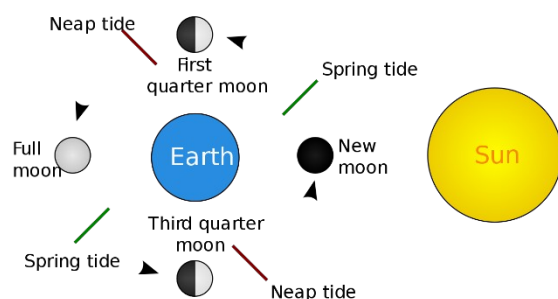
Newton's genius also gave us a **Third Law of Motion**, the **Law of Action and Reaction**, which states that

For every action, there is an equal and opposite reaction.

An action, or reaction, is a force impressed on some object.

If the Earth tugs on the Moon with a force of nearly 2×10^{20} Newtons, then the Moon (by the Third Law of Motion), must tug on the Earth with an equal but opposite force. This is true, and this reactive force gives rise to the water tides humankind sees every day.

The Sun also gravitationally tugs on the Earth (keeping Earth in orbit around the Sun), and so contributes to Earth's water tides too.



The gravitational force that the Moon exerts on the Earth stretches the Earth in such a way that the water (and to a lesser extent, the rocky surface) of the Earth is pulled outward along the line of sight between the Earth's and Moon's centers, both on the near and far sides of the Earth.

The Sun, to a lesser extent, also causes twice daily bulges of the Earth's watery surface. The combined action of the Moon's and Sun's gravitational tugs on Earth produces two high tides and two low tides daily.

If the Moon, Earth and Sun are lined up (at New Moon and Full Moon), the tides are highest, and are called **Spring Tides**. When the Sun and Moon are at right angles to the

Earth (at First Quarter Moon and Third Quarter Moon), the tides, while diminished, are still present, and are called **Neap Tides**.

In addition to little g being a function of height off the surface of the Earth, it is also a function of an observer's Earth latitude. Little g is at its maximum value at Earth's north and south poles, and is at its minimum value at Earth's equator.

This is because every object (not at the north or south pole) stuck to Earth by gravity makes a circle once per day, and this circular movement requires a small force, a force that must be supplied by the object's weight.

Objects at the equator make the biggest circles and experience the greatest reductions in their weights, whereas objects at the north and south poles make no circles and experience no reductions in their weights.

We see these reductions in nonpolar weights as reductions in the measured values of little g .

Little g is a function of latitude, with

g at the poles measuring 9.832 m/sec^2 , and

g at the equator measuring 9.780 m/sec^2 .

Latitudes between zero degrees (the equator) and plus or minus ninety degrees (the north and south poles respectively) have measured values of little g between 9.780 and 9.832.

The efficacy of little g being a constant is succinctly illustrated by a variety of practical problems.

For example, it is easy to show that for objects starting from rest and undergoing free fall,

$$s = \frac{1}{2} g t^2 \quad (3.7)$$

where s is the distance the object freely fell when let go, and t is the time it took for the object to fall that distance.



Consider now the plight of [Inspector Clouseau](#), who has recovered the [Regent Diamond](#) stolen from the [Louvre Museum](#) in Paris. In the inspector's usual bumbling fashion, he drops the precious stone into a Parisian sewer.

He knows he can climb down the ladder just below the sewer cover to retrieve the diamond, but he has a phobia about climbing down a ladder that is longer (taller) than 8 meters.

Using his iPhone stopwatch timer, the inspector drops a pebble through the sewer cover, and he hears it splash in the water below in 1.12 seconds.

Realizing that the time for the “splash sound” to go from the water back up to his ear (and phone) is very small (the speed of sound in air is 768 mph), he uses formula 3.7 to determine whether he can descend into the sewer and retrieve the diamond before a poor little Parisian mouse gobbles it up. Can he retrieve the diamond in time, or will his phobia win out?

We see that the maximum value for s is 8 m, and we have a known value of t of 1.12 sec. Using $g = 9.8 \text{ m/sec}^2$, we calculate the distance the diamond actually fell:

$$s = \frac{1}{2} (9.8) (1.12)^2 = 6.15 \text{ m (with rounding)}$$

Clearly it is safe for Inspector Clouseau to retrieve the diamond and receive the accolades he so richly deserves.



In the **First Extended Study** of Module 2A, we thoroughly studied the conflict that 16th century astronomers had regarding the true nature of the Solar System.

Was the Solar System (the Sun, the Moon and all the visible planets) moving around the Earth (the **geocentric model**), or was the Solar System (specifically all the visible planets) moving around the Sun (the **heliocentric model**)?

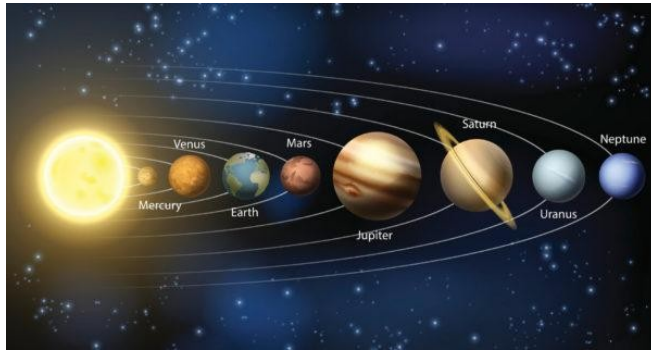
The polish cleric [Copernicus](#) inaugurated this conflict with his book "De Revolutionibus Orbium Coelestium" (On the Revolutions of the Celestial Spheres), which was first published in 1543 CE (as Copernicus lay dying).

Copernicus' model, while clearly simpler, had one flaw that initially inhibited its acceptance: Copernicus assumed that all planetary orbits around the Sun were circular in form. This is not true.

While Galileo Galilei, at his own peril, did much to promote the superiority of the Copernican model of the Solar System, it was a contemporary of Galileo, Johannes Kepler, that correctly fixed the Copernican model, and thus led to its general acceptance.

[Kepler](#) correctly asserted that the

orbits of planets were ellipses, not circles.



By **visible planets** is meant

all the planets (large spherical objects that orbit the Sun) that can be seen with the “naked eye” (without the aid of a telescope or binoculars).

In the picture to the left, we see the **visible planets** in order of their distance from the Sun, namely

Mercury, Venus, Earth, Mars, Jupiter, and Saturn.

The picture (not drawn to scale) above also shows two planets that can be seen only with the aid of a telescope, namely

Uranus and Neptune,

for a total of eight planets in our Solar System. Pluto is not included in the image above, as it has been (correctly) demoted to the status of a [dwarf planet](#).

Due to the way in which the Solar System formed, all of its planets formed within a thin disk of gas and dust, the center slice of which (for us) is

the plane of Earth's orbit around the Sun,

which is called **the Ecliptic Plane**. The orbit of the Earth itself is called **the Ecliptic**.

If you put an imaginary sphere of arbitrary radius around the Earth so that

its latitude and longitude are the same as Earth itself,

you have the **Celestial Sphere**.

The Celestial Sphere (basically the sky) can be divided up into

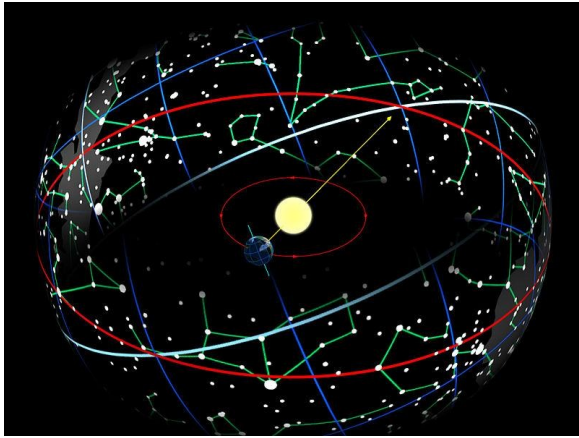
88 contiguous patches of sky

called the **constellations**.

Constellations that are in the region of

eight degrees latitude north and south of the Ecliptic Plane,

a region known as **the Zodiac**, are called the **zodiacal constellations**, of which there are thirteen.



Again, due to how the Solar System was formed,

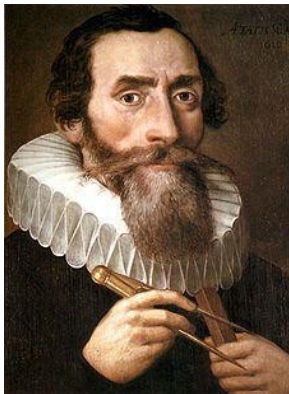
all the planets' orbits are in the Zodiac.

When it is convenient to think of the Earth as the center of the Solar System (for astronomical or astrological purposes), the Sun is visualized to

move around the Earth once per tropical year, traversing the zodiacal constellations.

Not only do the planets orbit the Sun, so do **other solar system objects**, notably
asteroids, comets, Kuiper Belt Objects, and the Oort Cloud.

We will study these other solar system objects extensively in a later module.



As mentioned above, Johannes Kepler “fixed” the Copernican model of the Solar System, thereby paving the way for the eventual total acceptance of [heliocentrism](#).

Kepler did so, as part of his [Three Laws of Planetary Motion](#), laws that he derived without theoretical foundation, but strictly from planetary observational data afforded him.

Kepler holds a unique and revered place in the history of astronomy. A contemporary of Galileo (his life was encapsulated by Galileo's), Kepler first worked for the “last of the great naked eye astronomers” [Tycho Brahe](#).

When Brahe died suddenly, Kepler was in the unique position to pour over Brahe's priceless data on Mars and other planets, and he was able, over an extended period of time, to determine his **Three Laws of Planetary Motion**.

Kepler's **First Law of Planetary Motion** states that

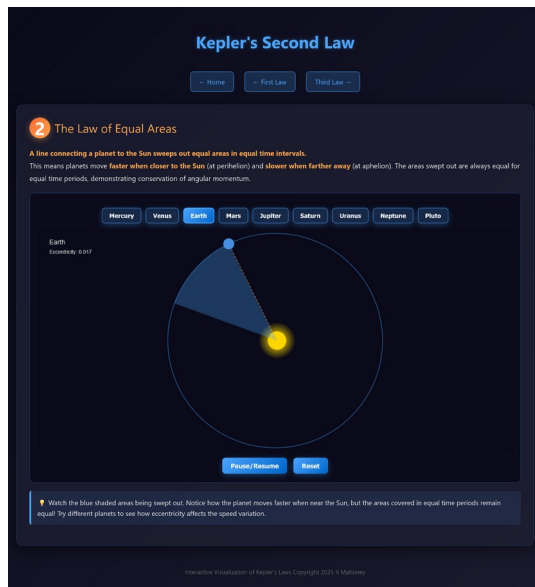
the orbit of a planet is an ellipse, with the Sun at one of the ellipse's two foci.

Think of an [ellipse](#) as “a squished circle” or oval.

An ellipse can be generated by tying a string between two pins (the foci), and holding a pencil tautly under the string so as to generate a series of points whose distances from the two pins always adds up to the same amount (the length of the string).

While there is only one Sun, there are eight planetary orbits, each having their own foci.

Kepler's **Second Law of Planetary Motion** is a bit more complicated. It states that



a line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time.

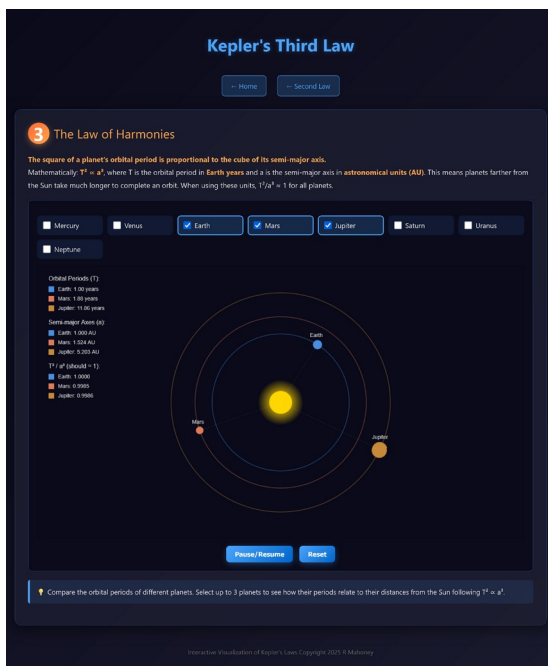
[Click](#) on the image to the left to see an animated version Kepler's Second Law of Planetary Motion.

<https://bobarebob.github.io/Animations/>

What's interesting about this second law, is that it explains how daylight is added and subtracted over the course of an Earth year. Because Earth must sweep out equal areas in equal times, when Earth is closer to the Sun (during the equinoxes, and during winter), it must move faster in its orbit.

This results in the daily increase in daylight time being greater just after Winter Solstice, than just before Summer Solstice.

Kepler's **Third Law of Planetary Motion** is the most obscure of the three, but probably of the most value of the three to astronomers. It states that

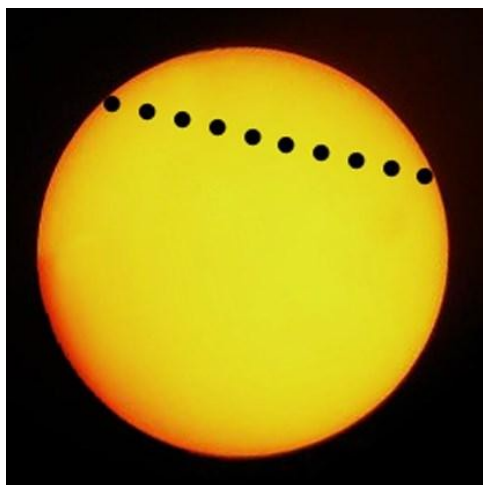


the square of the orbital period of a planet is directly proportional to the cube of the semi-major axis of its orbit.

Put in simple english, Kepler's Third Law says that the square of the duration of a planet's year (expressed in Earth years) is proportional to the cube of the average distance the planet is away from the Sun, in multiples of the average Earth-Sun distance (the astronomical unit or AU).

[Click](#) on the image above to see an animated version Kepler's Third Law of Planetary Motion.

Note: Newton was ultimately able to derive all of Kepler's Laws of Planetary Motion from his three laws of motion and his law of gravity.



Newton's good friend [Edmond Halley](#) suggested that the 1761 [transit of Venus](#) (the movement of the planet Venus across the face of the Sun, a very rare event) could be used to determine the average distance between the Sun and the Earth (the **astronomical unit** or AU, mentioned in Module 3A).

Numerous groups were sent to different locations on Earth to do this, and an estimate of 95 million miles was subsequently determined, a decent result compared to the current measurement of 93 million miles.

The determination of the AU allowed astronomers, using Kepler's Third Law of Planetary Motion, to subsequently determine the average distance to all the other eight planets, and thus to determine the size of the Solar System.

HOMEWORK

1. How Attractive Are You?

If gravity is an attractive force between any two masses, what is the gravitational attraction between you and your car?

Assuming you have a mass of 80 kg (corresponding to a weight of about 176 lbs), and assuming your car is a midsize sedan (corresponding to a curb weight of 3500 lbs, or having a mass of 1589 kg), and assuming that the distance between you and the center of your car is 1 m, the gravitational attraction between you and your car, from formula (3.4), would be

$$F = 6.67 \times 10^{-11} \times 1.589 \times 10^3 \times 80 / 1^2 = 8.479 \times 10^{-6} \text{ N}$$

which corresponds to the weight of a [small worker ant](#).

2. Why Doesn't the Earth Come Up to Meet You?



Let's say for the sake of argument that you have a mass of 100 kg. This corresponds to a weight of 980 N, or 220.3 lbs.

If you jump up about a meter, the Earth pulls you down with a force equal to your weight and your acceleration is about 9.8 m/sec^2 .

Newton's third law requires that you pull on the Earth with an equal and opposite force. Why doesn't the Earth come up to meet you half way?

Well, just because the Earth's force on you (your weight) is equal and opposite to your force on the Earth, that doesn't mean that the corresponding accelerations are equal.

The acceleration that you apply to Earth, $g_{\text{you on Earth}}$, is incredibly small:

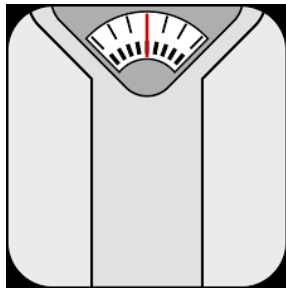
$$g_{\text{you on Earth}} = .164 \times 10^{-21} = 1.64 \times 10^{-22} \text{ m/sec}^2$$

From formula (3.7), with a fall distance (for you) of 1 meter. And a fall time (for you) of about .452 sec, it is not hard to show that the corresponding distance the Earth would travel up to meet you (d) would be

$$d = [1.64 \times 10^{-22} (.452)^2] / 2 = 16.72 \times 10^{-24} \text{ m}$$

This distance is about 17 trillionths of a trillionth of a meter, or about 10 billionths of the diameter of a hydrogen nucleus.

3. Weighing Yourself On The Moon



If you use formula (3.6) for determining the acceleration due to gravity on the Moon due to the Moon, we find that

$$g_{\text{Moon}} = 1.62 \text{ m/sec}^2 ,$$

Which is about 1/6th that of 9.8 m/sec², or 1/6th of g on Earth. It follows that if you weigh 180 lbs on Earth, you will only 30 lbs on the Moon.

Your mass will be the same, but, excluding your Moon gear, you will definitely be “lighter on your feet.”

4. The Earth Moon Distance



The Earth [was formed](#) about 4.54 billion years ago. [About 30 million years later](#), it was struck by a small planet or [large planetesimal](#) about the size of Mars, leading to the formation of the Moon from the subsequent ejecta.

Water accumulated on the Earth's surface [shortly thereafter](#), eventually covering two thirds of its surface. The Moon exerted a [tidal force](#) on the Earth's sluggish watery surface, stealing some of the Earth's rotational energy. Subsequently, the Moon gradually moved away

from Earth, and the Earth's day became longer.

At first the Moon was only about 20 to 30 thousand kilometers away from Earth, center to center, but over the course of 4.5 billion years, it migrated outward to its current average distance of 384,000 kilometers, or 239,000 miles, about 15 times farther away.

At first, the Earth day was somewhere between 4 and 6 hours, but now of course it is 24 hours. As little as 200 million years ago, when dinosaurs roamed our planet, the Earth day was only [23 hours long](#).

You can read more about this fascinating subject, and whether the elongation of Earth's day is in any way connected to the evolution of more complex life on Earth, by clicking [here](#).

5. See You in a Moon Day?

Even as the Moon has inexorably migrated away from Earth due to tidal forces, the rotational period of the Moon has slowed down to the point where the Moon rotates once on its axis as it revolves once around Earth. This [tidal locking](#) of the Moon's rotation has occurred over the billions of years that the Earth and Moon have been gravitationally connected.



The result is that one [Moon Day](#) and One Moon Year equals one Earth Month equals one lunation. This period is 29.5 days (a synodic month), if measured from New Moon to New Moon.

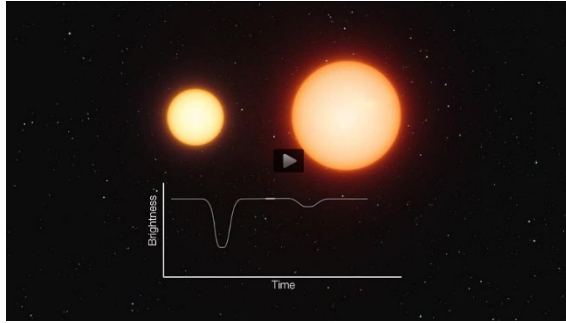
The Moon always shows only one side to us, its near side. We can see the near side of the Moon even at New Moon, because of light reflected off Earth and back at the Moon ([Earthshine](#)).

This does not mean that the other (nonvisible) side of the Moon, the far side, is dark, only that it is unobservable from Earth.

Due to many missions to the Moon, we now have many pictures of the far side of the Moon in various illuminations. For example, for New Moon here on Earth, the far side of the Moon will be fully illuminated.

[Click](#) on the image above to learn more about how the Moon is tidally locked with Earth.

6. Using Kepler's Third Law in Modern Astronomy



The constant of proportionality for Kepler's Third Law of Planetary Motion, so eagerly sought after by Edmond Hally and others, is related to the combined mass of the Sun and each planet orbiting the Sun.

But Kepler's Third Law [can be applied to any two objects circling each other](#), including binary (or two star) systems, which are quite common. If the two stars are of similar mass,

the period of revolution of one star around the other can be used to determine the two stars' joint mass, which divided by two, is the mass of each of the two stars.

[Click](#) on the image above to go to a Wikipedia article in which is embedded the video whose image is shown. When one star goes in front of the other star (eclipses it), the brightness of the star to the rear is diminished. This dimming action gives the astronomer enough information to determine the joint period of the binary star system and therefore the joint mass of the two star system.